Step4 Prompt in LCoT2Tree tool to assign function to each thought.

Your task is to classify Text2's purpose relative to Text1 using these categories:

Categories:
1. Continuous Logic - Direct continuation/extension of Text1's reasoning flow
2. Exploration - Introduces parallel/unrelated concepts from Text1, alternative reasoning paths, or new topics
3. Backtracking - Revises, corrects, or adjusts previous step
4. Validation - Provides supporting evidence, logical justification, or examples for Text1's claims

Input: {{
"Text1": "TEXT1",
"Text2": "TEXT2"
}}

Output Format:
Return only JSON format "'json{"Category": "Name of Category"}"'

Figure 8: The content of Step4 Prompt in LCoT2Tree tool to assign function to each thought.

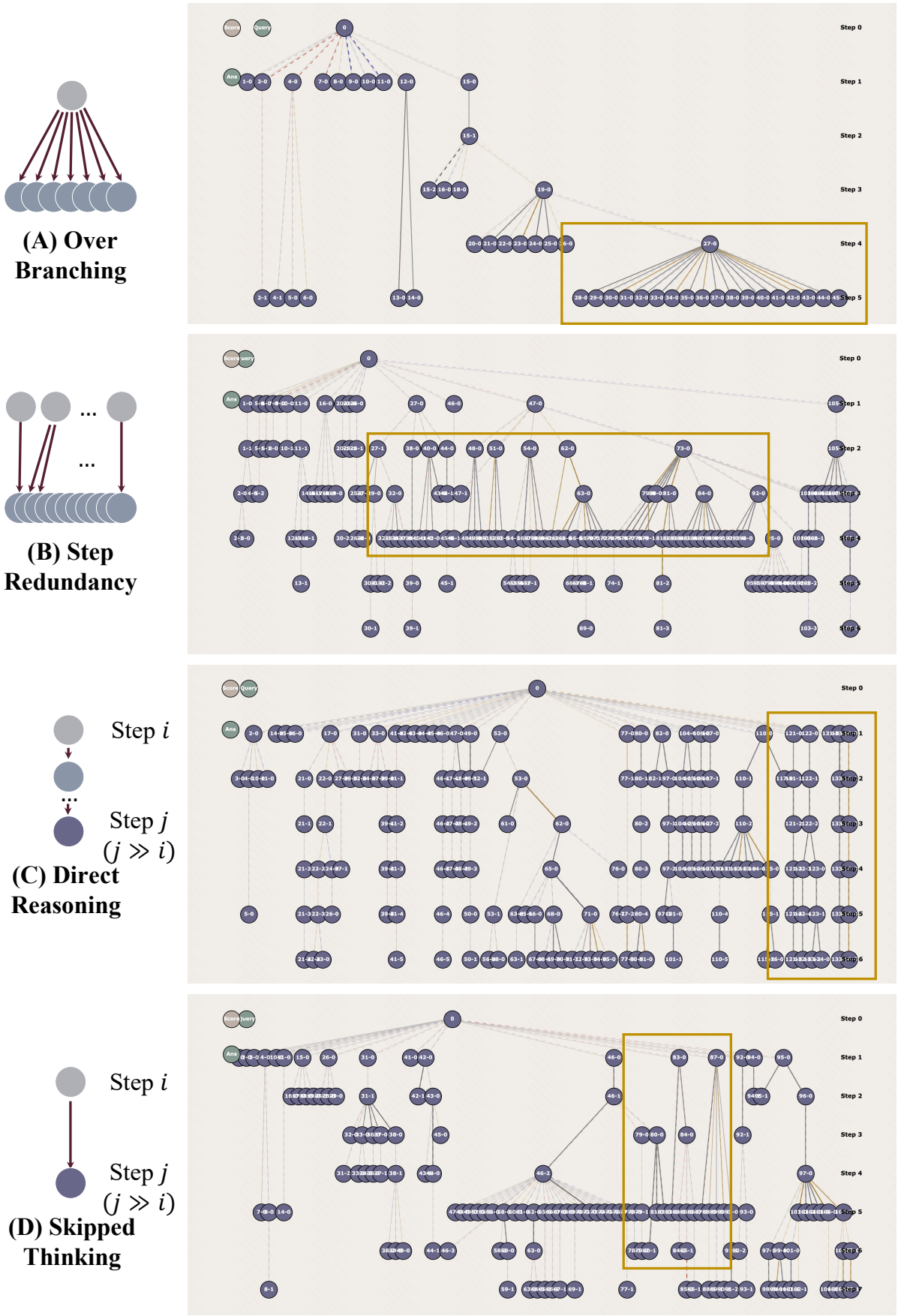

Figure 9: Visualization results of tree structure corresponding to different error patterns. The edge is labeled with the importance generated by GNNExplainer. The darker the color and the thicker the edge, the more important it is.

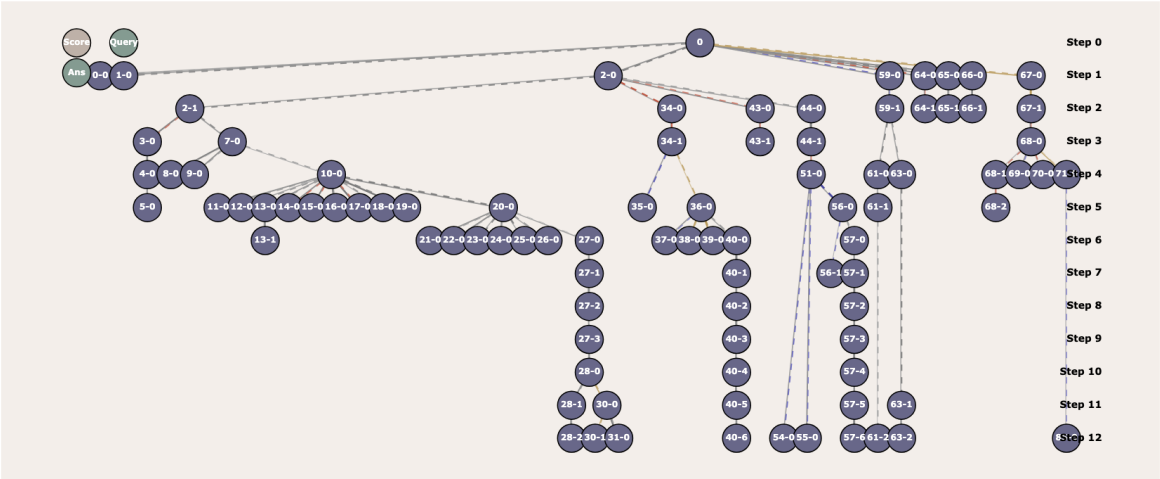

Figure 10: Visualization results of tree structure of a response from **DeepSeek-32B** on **MATH** dataset extracted using LCoT2Tree. The reasoning trees exhibit a downward-sloping hierarchical structure, with progressively deeper steps achieved through repeated backtracking.

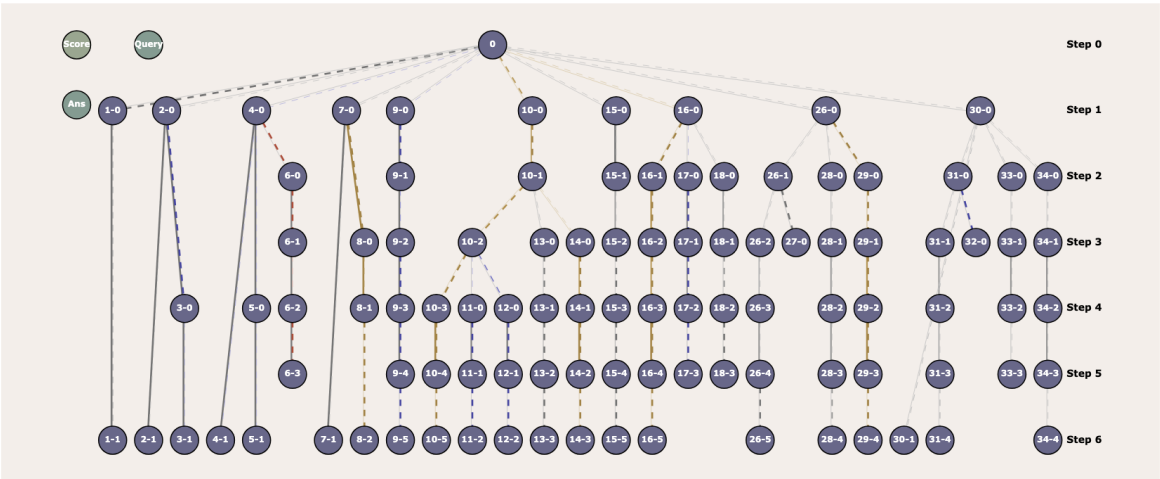

Figure 11: Visualization results of tree structure of a response from **DeepSeek-32B** on **LiveCodeBench** dataset extracted using LCoT2Tree. The reasoning patterns tend to show broad, parallel branching, where many sibling nodes initiate independent linear thought without subsequent exploration or verification.